

HAT-P-32b

R-0171

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_240106 · created 2026-07-16T17:47:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460315.7622 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.055
Transit depth (Rp/Rs)^2	3.3 +/- 2.0 [%]
Orbital Inclination (inc)	84.06 +/- 1.83
Airmass coefficient 1 (a1)	0.95 +/- 0.24
Airmass coefficient 2 (a2)	0.0285 +/- 0.0082
Scatter in the residuals of the lightcurve fit is	25.61 %
Best Comparison Star	#3 - [314, 291]
Optimal Aperture	2.63
Optimal Annulus	5.0
Transit Duration (day)	0.1208 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.055 vs 0.1489 (deviation 0.11σ)
Duration fit vs published	0.1208 d vs 0.1304 d (deviation 0.19σ)
Mid-time O–C	-10.3 min
Depth SNR	0.6
Residual scatter	25.61 %

Light curve

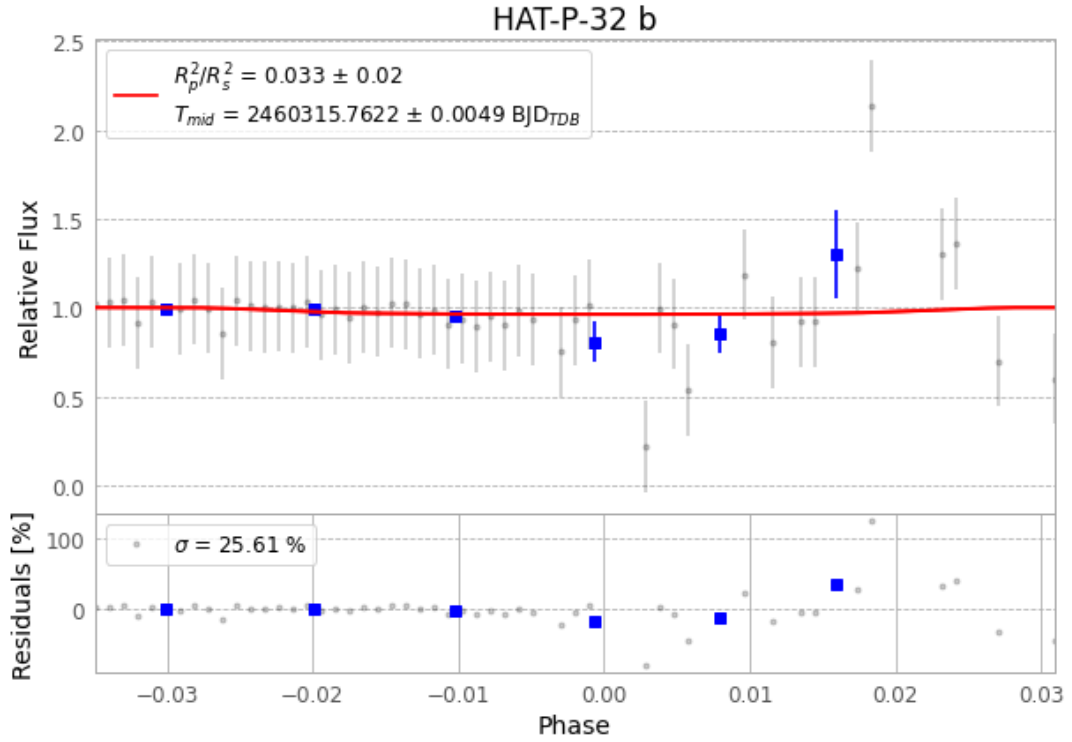


Figure 1 — FinalLightCurve_HAT-P-32 b_2024-01-05.png (SHA-256 c8f123f2be98d0f5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [321, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6227281c2ae97fad...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9b8a76a

Data provenance

70 FITS frames (46 MB), HATP-32240106042415.FITS → HATP-32240106075115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-240106.log (274d00a66412...), FinalLightCurve_HAT-P-32 b_2024-01-05.png (c8f123f2be98...), FinalLightCurve_HAT-P-32 b_2024-01-05.csv (8e77133592ca...)

Compute resources

Wall time 210.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 17:47Z.